The 0^{++} and 2^{++} glueball channels on center-vortex-removed SU(2) lattice ensembles: a data note

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Abstract

We measure zero-momentum glueball correlators on SU(2) Wilson-action ensembles at $\beta = 2.3$, 2.35, 2.4 and 2.5 in three readings of the same configurations: the original links, the center-projected Z_2 links in direct maximal center gauge, and the vortex-removed links of de Forcrand and D'Elia. A literature search found no published glueball measurement on vortex-removed ensembles; this is new as far as we found, not proven new. Removing the center vortices leaves a nonzero 0^{++} decay rate at $\beta = 2.35$, 2.4 and 2.5, but its value relative to the original glueball mass is not robust in these data. It moves with the gauge copy that defines the vortices: with the best of 8 copies at $\beta = 2.4$ (12^4) the vortex-removed mass rises from 0.653(42) to 0.836(78) in lattice units, a paired shift of 2.3σ, and with the best of 16 copies at $\beta = 2.5$ it also rises by 2.3σ. It moves with the coupling: the pre-registered ratio $m_{\mathrm{VR}}/m_{\mathrm{ORIG}}$ is 0.61(4) at $\beta = 2.35$ (10^4) against 0.78(3) at $\beta = 2.4$ (12^4), 3.4σ apart, and at $\beta = 2.4$ two independent data sets on the same box give 0.78(3) and 0.70(3). By the rule we fixed in advance the fraction is not robust, so we do not quote a single fraction as a property of the vortex-removed theory; in these data it depends on gauge fixing and coupling. What survives: the vortex-removed string tension is zero within errors at the largest distances we resolve at $\beta = 2.4$ and 2.5 (with a residual rise at intermediate distances, and a short-time estimator that fails at 2.5); at $\beta = 2.4$ the vortex-removed mass does not change between 12^4 and 16^4 (ratio 0.98(4), 5.6σ from the box ratio 0.75); and the vortex-removed 2^{++} level at $\beta = 2.4$, 1.231(63) in lattice units, lies 3.1σ above the lowest free two-gluon box level and falls only to 0.83(5) of the original 2^{++} mass while the 0^{++} falls to 0.70(3) on the same configurations, so the vortex-removed spectrum is not a uniformly scaled copy of the original one (pre-registered test, 2.4σ). Excited 0^{++} levels are upper bounds only, and at $\beta = 2.3$ the vortex-removed channel is not under control. Small lattices, SU(2) only, no continuum statement. We publish the measurements, the failure modes, data and code.

1. Limitations, stated first

- Monte Carlo evidence on small lattices: boxes from 8^4 to 20^4, at most two volumes per coupling in each volume test; $\beta = 2.35$ has one box (10^4) and no volume test.
- SU(2) only, Wilson action, four couplings ($\beta = 2.3, 2.35, 2.4, 2.5$), no continuum extrapolation. Only 2.35, 2.4 and 2.5 enter any vortex-removed mass comparison, because the $\beta = 2.3$ channel is not under control (sec. 4.2); at 2.35 the vortex-removed effective mass does not plateau either (sec. 4.5). Nothing here bears on the continuum mass-gap problem.
- The vortex-removed (VR) ensemble is gauge dependent: it is defined through direct maximal center gauge (MCG), which has many local maxima (Gribov copies). The main runs use one gauge copy per configuration, and every VR mass and ratio in this note is a single-copy value unless stated otherwise. Paired gauge-copy studies moved the VR mass by $+0.109(48)$ at $\beta = 2.5$ (best of 16, sec. 4.4), $+0.183(78)$ at $\beta = 2.4$ (best of 8, sec. 4.5) and $-0.192(109)$ at a $\beta = 2.3$ control, 17 to 28% of the mass; at 2.4 the pre-registered test calls the VR mass copy-sensitive. Copies closer to

the global maximum (simulated annealing, many copies [14]) and Laplacian center gauge [2] were not tested. The VR mass depends on the gauge copy in these data, and we do not know its value at the global maximum.

- VR configurations are not generated by a local reflection-positive action, so VR correlators need not be positive sums of exponentials; "mass" means the decay rate read by our estimator.
- Removing the vortices flips the sign of the trace on every P-plaquette, so the VR plaquette is much lower than the original (0.585 vs 0.630 at $\beta = 2.4$, 0.623 vs 0.652 at 2.5). VR configurations therefore carry a density of lattice-scale defects, and the VR 0^{++} channel could be partly a defect (ultraviolet) effect. The only checks here are that $m_{\mathrm{VR}}/\sqrt{\sigma}$ changes little between $\beta = 2.4$ and 2.5 ($2.52 \to 2.38$ in literature units) and the series 4 lattice-artifact test between 2.35 and 2.4, which is inconclusive (sec. 4.5).
- Estimator caveats: effective masses are resolved only to $t \approx 3$, and the VR effective masses are not constant there in every data set (at $\beta = 2.35$ they fall from $t = 2$; on the larger series 4 sample at $\beta = 2.4$, 12^4, $m(3)$ lies 2.7σ below $m(1)$), so a quoted VR mass, and any ratio built on it, depends on the time at which it is read (sec. 4.6); the volume tests use a fixed plateau rule on $m(1), m(2)$; the pre-registered string-tension premise check uses $V(R) = \ln W(R,3)/W(R,4)$, which probes short physical times and, with a free Coulomb term, can absorb part of a potential that rises and then flattens inside the fit window (sec. 4.3). With this estimator our units $a\sqrt{\sigma}$ differ from the literature values [9, 29] by factors 1.04 ($\beta = 2.3$, 16^4), 0.95 (2.3, $16^3 \times 32$), 1.05 (2.4) and 1.10 (2.5); we give masses in both units. In literature units the original glueball masses are 1 to 6% below the continuum value $3.78(7)\sqrt{\sigma}$ [10].
- Polyakov loops were not measured on the VR ensembles, so we cannot say whether center symmetry is broken there at $T = L$, which bears on which finite-volume states the 0^{++} operators can couple to (sec. 6).
- Not blind: the original-ensemble glueball mass was known in advance; the series 1 estimators were chosen after seeing exploratory 8^4 and 12^4 data, and those exploratory data are part of the series 1 sample (sec. 3); one deviation (GEVP t_0) was made after testing on already-seen data; the long-time estimators of sec. 4.3 and the gauge-copy method of sec. 4.4 were chosen after the $\beta = 2.5$ premise failure was known; the $\beta = 2.4$ long-time reanalysis was done after referee review; the series 4 rules were written with all series 1 to 3 results known, and two of its tests reuse series 1 configurations. The rules of each series were written and committed to a private git repository before that series' new production data existed (internally time-stamped, not externally registered).

2. Background: what is known and what is new

Center vortices as the confining configurations are reviewed in [17]. De Forcrand and D'Elia [1] removed center vortices from SU(2) configurations by multiplying each link by its center projection in direct MCG, and found that confinement is lost (Creutz ratios tend to zero, large-T slopes do not grow with R) at $\beta = 2.4$ on 16^4. Pepe and de Forcrand [2] found the same at $\beta = 2.3$ to 2.5 with Laplacian center gauge. Direct MCG and center dominance go back to Del Debbio et al. [3, 4]. Vortex removal has since been used for gluon and ghost propagators [5, 6, 7], the SU(2) quark propagator [30], the SU(3) string tension [31], and the light hadron spectrum and chiral symmetry in SU(3) [8], where it restores chiral symmetry. Biddle, Kamleh and Leinweber [16] found that the vortex-removed static potential is not described by

$V_0 + \sigma R - e/R$ at long range and fitted it with a modified Coulomb term. Glueballs after Abelian projection were measured in [11, 12]. Langfeld and Schafke [13] measured 0^+ and 2^+ screening masses on thick-vortex ensembles obtained by gauge-invariant cooling, which is a different reduction from the one used here. The dependence of center-projected observables on the MCG gauge copy is documented in [14, 15]: better maxima lower the projected string tension, and the effect weakens on larger volumes.

New here, as far as our search found (arXiv, INSPIRE including the papers citing [1], 1999 to October 2026): (a) the 0^{++} glueball correlator on VR ensembles and on thin Z_2 CP ensembles, with volume tests; (b) a quantification of how a short-time string-tension estimator misreads the VR potential (a residual of about 0.2 at $\beta = 2.5$ that long-time estimators do not show); the phenomenon itself is expected from [1, 2], which used long-distance observables, and from the failure of the standard potential form reported in [16]; (c) a paired gauge-copy study of that residual and of the VR mass; (d) the 2^{++} channel and the first excited 0^{++} level on VR ensembles, with a test of whether the VR spectrum is the original one scaled down.

3. Method

Ensembles. SU(2) Wilson action, heat bath [20, 21] with four over-relaxation [22, 23] steps per compound sweep, hot start, 300 thermalization sweeps. Four series, each with independent streams stopped by a wall clock fixed before any of its data were analysed:

- Series 1: $\beta = 2.3$ on 8^4, 12^4, 16^4 and $\beta = 2.4$ on 12^4, 16^4, $T = L$, a measurement every 2 sweeps. Part of this sample existed before the series 1 rules were written: the $\beta = 2.3$ 8^4 data (5000 measurements) and two of the three 12^4 streams at each coupling (1700 of 3670 measurements at $\beta = 2.3$, 1750 of 3730 at $\beta = 2.4$) come from earlier exploratory streams with the same action and gauge fixing, which had been analysed. New after the rules: one 12^4 stream per coupling and all 16^4 streams (five at 2.3, four at 2.4; three of them were added about 2.4 h after launch with the same deadline, before any data were analysed).
- Series 2: $\beta = 2.3$ on $12^3 \times 32$ and $16^3 \times 32$; $\beta = 2.5$ on 16^4 and 20^4; a measurement every 2 sweeps. All data new after the rules.
- Series 3 (gauge-copy study): $\beta = 2.5$ on 16^4 (8 streams) and 20^4 (2 streams), $\beta = 2.3$ on $12^3 \times 32$ (2 streams, control); a measurement every 10 sweeps. All data new after the rules.
- Series 4 (spectrum, gauge copies at 2.4, a second coupling): $\beta = 2.4$ on 12^4 (seed 701 with paired gauge copies, seed 702 single copy) and $\beta = 2.35$ on 10^4 (seeds 711, 712, single copy), a measurement every 2 sweeps, all four streams stopped at the same wall clock (launch + 2.5 h). All data new after the rules. It also reanalyses the series 1 $\beta = 2.4$ configurations (12^4, 16^4) for the excited 0^{++} level. At $\beta = 2.35$ no literature string tension was used: the smearing scale $a\sqrt{\sigma} = 0.313$ is the geometric mean of the 2.3 and 2.4 table values (our E1 gives 0.318(3)), and $10\, a\sqrt{\sigma} \approx 3.2$ makes the 10^4 box roughly the same physical size as 12^4 at 2.4 (3.35).

Center projection and vortex removal. Direct MCG maximizes $\sum_{x,\mu} |\mathrm{tr}\, U_\mu(x)|^2$ by local exact maximization with over-relaxation ($\omega = 1.85$) until the relative change is below 10^{-8}. With $Z_\mu(x) = \mathrm{sign\,tr}\, U_\mu(x)$, the CP links are $Z_\mu(x)$ and the VR links are $Z_\mu(x)U_\mu(x)$ [1]. On every plaquette of every configuration we assert $P_{\mathrm{orig}} = Z_P P_{\mathrm{VR}}$, which fails if the gauge fixing is not a gauge transformation.

Series 1 and 2 use one gauge copy (over-relaxation started from the configuration as generated). Series 3 fixes every configuration twice: the same single copy ("single"), and the best functional among 16 candidates, the single copy plus 15 Haar-random gauge transforms each relaxed to convergence ("best of 16"). Both are measured on the same configuration, so differences are paired. Series 4 does the same at $\beta = 2.4$ with the best of 8 candidates (the single copy plus 7 random copies).

Glueball. Zero-momentum sums of spatial Wilson loops [28] (1×1, 2×2, 3×3, 4×4, 1×2, 2×3, 3×4) at three APE smearing levels [24] ($\alpha = 0.5$; $(0, 2, 8)$ steps at $\beta = 2.3$, $(0, 4, 15)$ at 2.4, $(0, 8, 31)$ at 2.5, scaled with $(a\sqrt{\sigma}\,)^{-2}$): basis A, 21 operators (CP: the 7 shapes unsmeared). Series 4 stores the loop sums plane by plane: the 0^{++} (A_1^{++}) operator is the sum over the three spatial planes (asserted equal to the series 2 operator), the 2^{++} (E^{++}) operators are $(P_{xy} - P_{xz})/\sqrt{2}$ and $(P_{xy} + P_{xz} - 2P_{yz})/\sqrt{6}$, both components pooled in one 21×21 correlator matrix without vacuum subtraction, and the first excited 0^{++} level is the second GEVP vector at the same (t_0, t). Smearing levels $(0, 3, 11)$ at $\beta = 2.35$. Vacuum-subtracted correlators averaged over all time slices, normalized by $\sqrt{\mathrm{diag}\, C(0)}$, ground vector from the generalized eigenvalue problem [25, 26] at $(t_0, t) = (0, 1)$, effective masses $m(t) = \ln C(t)/C(t + 1)$. Plateau rule, fixed in advance: if $m(2)$ has relative error below 30% and $|m(1) - m(2)| < 1.5\sigma$ (jackknife of the difference), quote $m(1)$ ("plateau"); if $m(2)$ is that precise but differs, quote $m(2)$ ("late"); otherwise quote $m(1)$ as an "upper bound"; a quoted value with relative error above 30% is "noisy".

String tension. Estimator E1: $V(R) = \ln W(R, 3)/W(R, 4)$, fitted with $V_0 + \sigma R - e/R$, e free, $R = 2, \ldots, L/2$, with spatially smeared Wilson loops. Series 3 and the reanalyses of sec. 4.3 add E1′ (the same fit on $R \geq 4$), E2 (the same fit on $\ln W(R, T)/W(R, T + 1)$ at $T = 5$ and $T = 6$) and Creutz ratios $\chi(R, R)$. Ratios $\sigma_X/\sigma_{\mathrm{ORIG}}$ use the same estimator on the same configurations. Literature values used for units and checks: $a\sqrt{\sigma}\,$ = 0.369 ($\beta = 2.3$), 0.266 (2.4), 0.188 (2.5) [9, Table 10; 29, Table 7].

Errors. Delete-one-block jackknife, block length at least twice the largest integrated autocorrelation time of the plaquette and the 2×2 operator (Madras-Sokal window [27], $c = 6$), 8 to 20 blocks per stream; the bin-size scan of the VR mass errors (4 to 40 blocks per stream) is in data/bin_size_scan.csv. Autocorrelation times of the gauge-fixed observables themselves were not estimated. Thermalization check: first quarter dropped if its plaquette differs from the rest by more than 3σ (one stream in series 1).

Pre-registered tests. Premise C0: $\sigma_{\mathrm{VR}}/\sigma_{\mathrm{ORIG}} < 0.15$ and $\sigma_{\mathrm{CP}}/\sigma_{\mathrm{ORIG}} \in [0.6, 1.4]$ with E1. Volume test at each coupling: $\rho = m(L_{\mathrm{large}})/m(L_{\mathrm{small}})$ in lattice units; "gap" if ρ is within 2σ of 1 and more than 2σ from the box ratio $L_{\mathrm{small}}/L_{\mathrm{large}}$, "gapless" the reverse, otherwise inconclusive; interpreted only if the original-ensemble control reads "gap". The series 4 tests are listed in sec. 4.5.

4. Results

4.1 Statistics and checks

Plaquettes: 0.60228 ($\beta = 2.3$), 0.63001 (2.4), 0.6519 to 0.6520 (2.5). At $\beta = 2.3$ and 2.5 they agree within 3×10^{-4} with [29, Table 7] (0.6025, 0.6520); the same table lists $1 - 0.36352 = 0.6365$ at 2.4, which does not lie between its neighbours and which we could not confirm elsewhere; our 0.63001 does. Integrated

autocorrelation times are 0.5 to 1.6 measurements (one short stream 3.0). No assertion fired in any stream. MCG functional: 0.753 (2.3), 0.773 (2.4), 0.791 (2.5, single copy). P-vortex density: 0.090, 0.055, 0.032.

4.2 Masses

Table 1. Basis A, lattice units, quoted mass with plateau flag (pl = plateau). VR effective masses $m(1), m(2), m(3)$ in the last column.

β	lattice	N	ORIG	VR	CP	VR $m(1), m(2), m(3)$
2.3	8^4	5000	1.134(54) pl	0.719(83) late	0.863(24) pl	1.022(41), 0.719(83), 0.226(88)
2.3	12^4	3670	0.859(104) late	0.943(28) pl	0.863(19) pl	0.943(28), 0.944(93), 0.458(121)
2.3	16^4	1228	1.049(79) pl	0.691(103) late	0.847(26) pl	1.018(57), 0.691(103), 0.481(158)
2.3	$12^3 \times 32$	1510	1.216(42) pl	1.052(43) pl	0.880(20) pl	1.052(43), 1.004(126), 0.433(126)
2.3	$16^3 \times 32$	840	1.325(67) pl	0.891(104) late	0.926(26) pl	1.050(42), 0.891(104), 0.508(142)
2.4	12^4	3730	0.873(25) pl	0.680(18) pl	0.609(17) late	0.680(18), 0.688(34), 0.548(56)
2.4	16^4	1230	0.942(47) pl	0.669(22) pl	0.640(28) late	0.669(22), 0.661(51), 0.577(91)
2.5	16^4	920	0.709(36) pl	0.475(18) pl	0.419(19) late	0.475(18), 0.467(32), 0.509(56)
2.5	20^4	560	0.701(36) pl	0.448(21) pl	0.371(22) late	0.448(21), 0.416(34), 0.436(56)

Units $a\sqrt{\sigma_{\mathrm{ORIG}}}$ (E1, largest box at each coupling): 0.383(26) (2.3, 16^4), 0.350(18) (2.3, $16^3 \times 32$), 0.2789(34) (2.4), 0.2072(10) (2.5), against the literature 0.369, 0.266, 0.188 [9, 29]. At $\beta = 2.5$ the E2 ($T = 5$) estimator on the original 16^4 links gives $a^2\sigma = 0.0359(21)$, $a\sqrt{\sigma} = 0.189(6)$, the literature value, so the bias is in E1. In literature units $m_{\mathrm{ORIG}}/\sqrt{\sigma} = 3.54(18)$ (2.4, 16^4) and 3.73(20) (2.5, 20^4). For comparison only, the lowest non-zero-momentum free two-gluon level is $4\sin(\pi/L)$: 1.035 ($L = 12$), 0.780 (16), 0.626 (20); it is not the lowest free-field level in a periodic box (sec. 6).

Table 2. Volume tests (basis A) and ratios at the largest box. Last column: own E1 unit / literature unit.

β	**boxes**	ρ_{VR}	ρ_{ORIG}	ρ_{CP}	**VR call**	$m_{\mathrm{VR}}/m_{\mathrm{ORIG}}$	$m_{\mathrm{VR}}/\sqrt{\sigma}$
2.3	$16^4/12^4$	0.733(111)	1.22(18)	0.981(37)	gapless	0.66(10)	1.80(30) / 1.87(28)
2.3	$16^3\times 32/12^3\times 32$	0.847(105)	1.089(67)	1.053(38)	inconclusive	0.673(83)	2.55(33) / 2.42(28)
2.4	$16^4/12^4$	0.984(42)	1.079(62)	1.051(55)	gap	0.711(36)	2.40(8) / 2.52(8)
2.5	$20^4/16^4$	0.944(57)	0.989(72)	0.886(66)	gap	0.639(43)	2.16(10) / 2.38(12)

At $\beta = 2.3$ the VR effective mass falls with t on 8^4, 16^4 and $16^3\times 32$ and holds only on the 12 boxes; the plateau rule then quotes $m(2)$ on one box and $m(1)$ on the other, which drives the "gapless" call of series 1 and the inconclusive call of series 2. The $\beta = 2.3$ VR channel is not under control. At $\beta = 2.4$ every VR effective mass plateaus; the plateau rule quotes $m(1)$ on both boxes, and with $m(2)$ on both boxes the volume ratio is 0.96(9) (report only, computed after referee review). On the larger series 4 sample at 12^4 the VR $m(3)$ lies 2.7σ below $m(1)$ (sec. 4.6), so this plateau holds to $t = 2$. At $\beta = 2.5$ $m(1) \approx m(2) \approx m(3)$ in both boxes.

4.3 String tension and the premise check

E1 ratios $\sigma_{\mathrm{VR}}/\sigma_{\mathrm{ORIG}}$: $-0.09(4)$, $-0.04(1)$, $-0.02(1)$ ($\beta = 2.3$; $L = 8, 12, 16$), $-0.048(8)$, $-0.024(10)$ (2.3, $T = 32$), $-0.015(8)$, $+0.016(8)$ (2.4), but 0.219(10) (16^4) and 0.205(7) (20^4) at 2.5: C0 fails at $\beta = 2.5$. CP ratios are 0.94 to 1.17 everywhere (series 1 and 2).

Table 3. Long-time estimators, ratio VR/ORIG, single gauge copy. The $\beta = 2.5$ rows (series 2 configurations) were computed after the C0 failure; the $\beta = 2.4$ rows (series 1 configurations, the stored Wilson loops) were computed after referee review with the same estimator code. At 12^4, $T \le 6$ leaves no E2 at $T = 6$ and no $\chi(7,7)$.

β**, lattice**	**E1**	**E1′** ($R \ge 4$)	**E2,** $T = 5$	**E2,** $T = 6$	$\chi(5,5)$	$\chi(6,6)$	$\chi(7,7)$
2.4, 12^4	$-0.015(8)$	0.079(49)	$-0.174(42)$		0.126(20)	$-0.017(45)$	
2.4, 16^4	0.016(8)	0.060(22)	$-0.113(49)$	$-0.083(39)$	0.116(19)	$-0.004(63)$	$-0.017(81)$
2.5, 16^4	0.219(10)	0.191(25)	$-0.079(29)$	$-0.190(62)$	0.429(24)	0.207(56)	0.062(101)
2.5, 20^4	0.205(7)	0.213(13)	$-0.046(21)$	$-0.029(52)$	0.397(19)	0.237(50)	0.044(80)

The VR Creutz ratios (relative to ORIG) at $\beta = 2.5$, 16^4 fall $0.97, 0.83, 0.63, 0.43, 0.21, 0.06$ for $R = 2$ to 7 ; at $\beta = 2.4$, 16^4 they fall $0.86, 0.56, 0.28, 0.12, 0.00, -0.02$; at $\beta = 2.3$ ($16^3\times 32$) they are $0.68, 0.27, 0.07$ for $R = 2$ to 4 and zero within errors beyond. The VR potential rises at short distance and is flat within errors beyond $R \approx 4a$ (2.3), $6a$ (2.4) and $7a$ (2.5), roughly 1.4 to $1.7/\sqrt{\sigma}$ at each coupling. E1 fits $T = 3 \to 4$ over $R = 2$ to $L/2$ with a free Coulomb term, so it is dominated by the rising intermediate-distance part when the flattening lies well inside the window (2.5), while at 2.4 the free e absorbs part of

the rise and E1 reads zero. The negative E2 values are a fit artifact of $V_0 + \sigma R - e/R$ with free e on a potential that rises and then flattens, not a negative string tension; the same functional form fails on VR potentials in [16].

What this means for the $\beta = 2.4$ premise (analysis done after referee review, not pre-registered, reported because it bears on the headline coupling): every long-time ratio at 2.4 is below the pre-registered threshold 0.15, and the VR string tension is consistent with zero at the largest distances we resolve ($\chi(6,6)$, $\chi(7,7)$, $R = 6, 7$, about 1.6 to $2.0/\sqrt{\sigma}$). It is not zero at intermediate distances ($\chi(5,5) = 0.116(19)$, E1$' = 0.060(22)$). So the statement "the VR string tension is zero within errors" at 2.4 holds for E1 and at long range, not at all distances; the 2.4 and 2.5 ensembles differ mainly in where the flattening falls relative to the E1 window. Item (b) of sec. 2 is this quantification.

4.4 Gauge-copy study (series 3)

Rules were fixed before the series 3 data existed. The best-of-16 functional exceeds the single copy on 94% of the 668 16^4 configurations at $\beta = 2.5$, by 0.00215(5) on average (0.79115 → 0.79330). For comparison, [14] reports a mean maximal functional of about 0.797 to 0.798 on the same lattice (16^4, $\beta = 2.5$) with simulated annealing and many copies, so our best-of-16 copies are not close to the global maximum. The single-copy E1 ratio 0.205(9) reproduces series 2 (0.219(10)) on independent configurations, and the $\beta = 2.3$ control passes ($-0.046(21)$).

Table 4. Series 3, 16^4, $\beta = 2.5$, $N = 668$: string-tension ratios to ORIG, single copy vs best of 16 (paired difference).

estimator	VR single	VR best of 16	VR diff.	CP single	CP best of 16	CP diff.
E1	0.205(9)	0.225(10)	+0.020(13)	0.985(19)	0.726(14)	−0.259(12)
E2, $T = 5$	−0.042(37)	−0.044(37)	−0.002(50)	1.086(70)	0.765(50)	−0.321(28)
E2, $T = 6$	−0.045(95)	−0.151(109)	−0.106(135)	1.70(33)	1.16(22)	−0.54(11)
$\chi(6,6)$	0.232(72)	0.357(66)	+0.125(83)	1.013(94)	0.728(67)	−0.285(31)
$\chi(7,7)$	0.188(217)	0.106(200)	−0.082(285)	1.35(51)	0.93(35)	−0.42(16)

Ranking the 16 candidates by functional, the CP E1 ratio falls almost monotonically from 1.32 (worst) to 0.73 (best), while the VR E1 ratio rises mildly from 0.17 to 0.23. At $\beta = 2.5$, 16^4, the VR glueball mass (basis A) moves from 0.451(22) (plateau) with the single copy to 0.561(44) (late) with the best copy, a paired difference of +0.109(48), 2.3σ ($m(1)$: +0.051(28)); the single-copy value agrees with series 2 (0.475(18)). At the $\beta = 2.3$ control ($12^3 \times 32$, 241 configurations) the VR mass moves the other way, from 1.152(106) to 0.961(60) (both plateau), a paired difference of −0.192(109), -1.8σ. The 20^4 sample at $\beta = 2.5$ (54 configurations) gives E1 VR ratios 0.223(17) (single) and 0.252(23) (best), CP 0.970(22) and 0.754(22). It is too small for masses: its single-copy VR mass comes out 0.300(74) ("late"), against 0.448(21) in series 2.

4.5 Series 4: excited and 2^{++} levels, gauge copies at $\beta = 2.4$, and $\beta = 2.35$

Pre-registration status. The series 4 rules (readings, predictions, kill criteria, combined rule, expected outcome) were written and committed before any series 4 production data existed and before any excited-level, 2^{++} or $\beta = 2.35$ analysis on any ensemble, but after all series 1 to 3 results were known; internally time-stamped, not externally registered. The clock time written in the rules' header was wrong and was corrected in the launch log before any data existed (the text was written in the 13 minutes before launch). The four streams first died before their first checkpoint with a serialization error (one module loaded twice, so the lattice object could not be saved). The fix was one line with no change to the physics; the self-test was re-run and all four streams were restarted from scratch about 1.5 minutes later with the same deadline. No data from the first launch existed. The analysis code was written after launch and implements the written rules. The written expectation was that the VR mass would be copy-stable at 2.4 and that the fraction would scale; both came out the other way.

Data sets and tests. S1: the series 1 $\beta = 2.4$ configurations (12^4, $N = 3730$; 16^4, $N = 1230$; 0^{++} operators only). D-24: $\beta = 2.4$, 12^4, both new streams, single copy ($N = 3237$). D-copy: the paired stream ($N = 457$). D-235: $\beta = 2.35$, 10^4 ($N = 13126$). Errors: delete-one-block jackknife, 20 blocks per stream, ratios on aligned blocks, differences between independent data sets in quadrature. The tests, with 2σ thresholds, as written:

- G24 (gauge copies at 2.4, D-copy): the paired difference of the quoted VR 0^{++} mass, best of 8 minus single, is "copy-stable" below 2σ and "copy-sensitive" above; tested only if the best copy beats the single copy on at least 80% of configurations.
- R3 (the fraction scales): $f = m_{\mathrm{VR}}/m_{\mathrm{ORIG}}$ at $\beta = 2.35$ (10^4) equals f at 2.4 (S1, 12^4, the matched box): "holds"; otherwise "killed". Premise at 2.35: C0 with E1 passes and both quoted masses are flagged plateau or late.
- R2 (lattice-unit artifact): $\rho_a = am_{\mathrm{VR}}(2.35)/am_{\mathrm{VR}}(2.4, 12^4)$ against 1 (artifact) and against $s = a\sqrt{\sigma}\ (2.35)/a\sqrt{\sigma}\ (2.4)$ (physical scaling).
- R1 (one attractive short-range well binding two gluons in an S wave, with only a two-gluon continuum above): killed if the VR excited 0^{++} is volume stable (ratio $16^4/12^4$ within 2σ of 1 and more than 2σ from the free box ratio 0.754), or if a separate bound 2^{++} lies above the VR 0^{++} but more than 2σ below the box level $4\sin(\pi/L)$.
- R4 (scaled tower): for the excited 0^{++} (S1) and the 2^{++} (D-24), $m_X(\mathrm{VR})/m_X(\mathrm{ORIG})$ equals the ground-state ratio on the same data within 2σ; killed if any tested level differs by more.
- Combined: the fraction is "robust (in this window)" if R3 holds and G24 is copy-stable, "not robust" if R3 is killed or G24 is copy-sensitive.

Checks. Plaquettes 0.63004 (2.4, D-24) and 0.61691 (2.35); integrated autocorrelation times at most 1.0 measurement; no assertion fired. E1 gives $a\sqrt{\sigma} = 0.2787(35)$ (D-24) and 0.3184(33) (2.35). E1 ratios VR/ORIG: $-0.017(9)$ (D-24), $-0.076(9)$ (2.35); CP/ORIG: 0.973(27), 1.026(21); the C0 premise at 2.35 passes. In D-copy the best copy beats the single copy on 88% of configurations (mean functional gain

0.00198(7)), so G24 is tested. On D-24 the original-link masses in our E1 unit are $3.38(14)_{\sqrt{\sigma}}$ (0^{++}) and $5.33(24)_{\sqrt{\sigma}}$ (2^{++}), against the continuum 3.78(7) and 5.45(11) [10].

Table 5. Series 4 masses, basis A, lattice units, quoted by the plateau rule (pl = plateau, ub = upper bound). Last column: $m_{\mathrm{VR}}/m_{\mathrm{ORIG}}$ (0^{++} ground state, jackknife on aligned blocks). All rows single copy except "best of 8".

data set	**β, lattice**	**ORIG 0^{++}**	**VR 0^{++}**	**VR 0^{++} level 1**	**ORIG 2^{++}**	**VR 2^{++}**	**ratio**
S1	2.4, 12^4	0.873(25) pl	0.680(18) pl	1.755(214) ub			0.779(32)
S1	2.4, 16^4	0.942(47) pl	0.669(22) pl	1.877(333) ub			0.711(36)
D-24	2.4, 12^4	0.943(36) pl	0.661(16) pl	1.921(266) ub	1.486(63) pl	1.231(63) pl	0.701(34)
D-copy, single	2.4, 12^4	0.959(87) ub	0.653(42) pl	noisy			0.681(67)
D-copy, best of 8	2.4, 12^4	(same)	0.836(78) pl	0.981(175) ub			0.872(98)
D-235	2.35, 10^4	1.190(82) late	0.724(30) late	2.385(386) ub	1.321(271) late	1.469(57) pl	0.609(38)

The original excited 0^{++} levels (S1) are 1.673(162) and 1.399(185), both upper bounds. All effective masses and both levels per channel are in data/series4_summary.csv; the full analysis output, including every ratio below, is in data/series4_analyze.log.

Table 6. Series 4 tests, rules applied as written.

test	**measured**	**verdict**
G24, gauge copies	best of 8 minus single: +0.183(78), 2.3σ (ratio 0.681(67) → 0.872(98))	copy-sensitive
R3, fraction scales	$f(2.35) = 0.609(38)$ vs $f(2.4) = 0.779(32)$: $-0.171(50)$, 3.4σ	killed
R2, lattice artifact	$\rho_a = 1.065(52)$, $s = 1.153(17)$: 1.2σ from 1, 1.6σ from s	inconclusive
R1, excited 0^{++}	$\rho^* = 1.07(23)$ (0.3σ from 1, 1.4σ from 0.754), both upper bounds	inconclusive
R1, 2^{++}	VR 2^{++} 1.231(63), 3.1σ above 1.035; +0.570(62) above the VR 0^{++}	not killed
R4, 2^{++}	ratio 0.829(49) vs 0^{++} 0.701(34): +0.128(53), 2.4σ	killed
R4, excited 0^{++}	+0.270(178) (12^4), +0.631(292) (16^4), upper bounds	killed at 16^4 as written (2.2σ, borderline)
combined	R3 killed, G24 copy-sensitive	**fraction not robust**

Also measured (not part of a pre-registered decision): at $\beta = 2.35$ the VR 2^{++} is 4.1σ above its box level 1.236, and its ratio to the original 2^{++} exceeds the 0^{++} ratio by +0.503(228) (2.2σ; the original 2^{++}

there is only a late value with a large error). In D-copy the CP E1 ratio drops from 1.061(56) to 0.873(42) with the best of 8 (paired $-0.188(30)$, 6σ), as at $\beta = 2.5$, while the VR E1 ratio stays consistent with zero ($-0.059(23)$ single, $-0.006(28)$ best): at 2.4 the copy shift of the VR mass happens with no VR string tension in E1, unlike 2.5, where a residual was present.

4.6 Post-hoc observations on series 4 (not pre-registered)

These were seen after the series 4 data and were not part of its rules.

1. The R3 kill depends on the time at which the fraction is read. At $\beta = 2.35$ both quoted masses are "late" ($m(2)$), and the VR effective mass keeps falling: 0.840(15), 0.724(30), 0.455(50), 0.225(42) for $t = 1$ to 4; it never plateaus. Same-t fractions (jackknife): at $t = 1$, 0.789(19) at 2.35 against 0.779(32) at 2.4 (agree); at $t = 2$, 0.609(38) against 0.835(66). At 2.4 the D-24 VR $m(3) = 0.531(45)$ is 2.7σ below $m(1) = 0.661(16)$, so the single-exponential behaviour at 2.4 holds to $t = 2$ only.
2. The two $\beta = 2.4$, 12^4 data sets disagree at 1.6σ on the original mass (0.943(36) in D-24, 0.873(25) in S1), so the 12^4 fraction is 0.70(3) or 0.78(3) depending on the data set; with the 16^4 value 0.711(36) the band at 2.4 is about 0.70 to 0.78.
3. The VR 2^{++} lies near twice the VR 0^{++}: ratio 1.86(10) at 2.4 and 2.03(11) at 2.35. A reading of the VR 2^{++} as two VR scalars fits these numbers; it is untested.
4. The gauge-copy shift has the same sign and size in σ at 2.4 and 2.5 ($+2.3\sigma$ both). D-copy is small (457 configurations, one stream), and with several tests at 2.3 to 3.4σ one of them being a fluctuation is plausible.

5. Pre-registered outcomes

Each series fixed its combined rule in advance; we report them as written. Series 1 to 3 ask whether a gap survives vortex removal; series 4 asks whether the VR to original mass ratio is robust.

- Series 1: "gapless", triggered by $\beta = 2.3$ alone ($\rho_{\mathrm{VR}} = 0.733(111)$) while $\beta = 2.4$ was a "gap".
- Series 2: the combined rule as written reads "gap survives vortex removal (in this window)" ($\beta = 2.5$ gap, 2.4 gap, no significant drop of $m_{\mathrm{VR}}/\sqrt{\sigma}$ from 2.4 to 2.5, 2.3 not gapless). The rule did not say whether the series 1 consequence of a C0 failure (that coupling does not enter the verdict) carries over; with it, the outcome is inconclusive.
- Series 3 was designed to settle that by test. Its rule: if the best-of-16 VR E1 ratio is still ≥ 0.15, the combined outcome is inconclusive, because the $\beta = 2.5$ VR point is not a clean vortex-free measurement under C0. The ratio is 0.225(10), so the **combined outcome is inconclusive**. The pre-registered gauge-fixing diagnosis (G) does not hold: with best-of-16 copies the residual does not fall (it rises slightly with the functional). Copies closer to the global maximum (simulated annealing, [14]) and Laplacian center gauge [2] were not tested. "Measurement artifact of the short-time estimator" and "real residual" are not separated by the pre-registered test, because the long-range estimators at 16^4 have errors of about 0.2.

- Series 4: the **fraction is not robust**. The gauge-copy test at $\beta = 2.4$ is copy-sensitive (+0.183(78), 2.3σ) and the scaling test between $\beta = 2.35$ and 2.4 is killed (3.4σ); either alone decides it. The scaled-tower reading is killed by the 2^{++} (2.4σ); the short-range binding reading is not killed; the lattice-artifact test and the excited-level volume test are inconclusive (Table 6).

6. Discussion

What these data support: removing the center vortices in direct MCG leaves a nonzero 0^{++} decay rate at $\beta = 2.35$, 2.4 and 2.5, and removes the string tension at the largest distances we resolve at 2.4 and 2.5. The size of that decay rate relative to the original glueball mass is not a stable number in these data. It moves toward the original mass when the gauge copy is improved, by 2.3σ (24 to 28% of the single-copy mass) at both 2.4 and 2.5; between $\beta = 2.35$ and 2.4, in boxes of about the same physical size, the pre-registered ratio changes from 0.61(4) to 0.78(3); and two data sets on the same 2.4 box give 0.78(3) and 0.70(3). Read at $t = 1$, the 2.35 and 2.4 ratios agree (post hoc, sec. 4.6), so part of the coupling dependence is the time dependence of a VR correlator that is not a single exponential. The value 0.71(4) reported in earlier versions of this note ($\beta = 2.4$, 16^4) is therefore a single-copy, single-coupling, plateau-rule number, not a property of the vortex-removed theory. Whether a copy-independent and coupling-independent ratio exists, and what it is, is open.

The volume test at $\beta = 2.4$ still holds for the single-copy mass: it does not change between 12^4 and 16^4, $\rho = 0.984(42)$, 5.6σ from the ratio 0.75 that a level scaling as $1/L$ would give (0.96(9) with $m(2)$). This volume ratio is the evidence against a finite-volume state, and it has two limits. First, the free two-gluon level we compare with, $4\sin(\pi/L)$, is the lowest non-zero-momentum level; in a periodic box the lowest free-field 0^{++} levels lie in the zero-momentum (torus) sector, with energies of order $g^{2/3}/L$ at weak coupling [18, 19], which can be below it. Being below $4\sin(\pi/L)$ therefore does not by itself exclude a finite-volume state, and we keep that comparison as secondary only. Second, with $\sigma_{\mathrm{VR}} = 0$ at long range a pair of Polyakov loops (flux winding the box) has no energy cost linear in L, so a volume-independent level could also be such a finite-volume state; we did not measure Polyakov loops. The volume test uses one gauge copy (the paired test at the same coupling calls that mass copy-sensitive), and the string tension at 2.4 is zero within errors with E1 and at the largest distances, not at intermediate distances (sec. 4.3). The single-copy mass is $2.40(8)\sqrt{\sigma}$ in our own unit and $2.52(8)\sqrt{\sigma}$ in the literature unit.

The 2^{++} channel separates two readings of the surviving state. A vortex-removed spectrum that is the original tower scaled by one factor is killed by the 2^{++} (pre-registered, 2.4σ): on the same configurations the VR 2^{++} falls to 0.83(5) of the original and the 0^{++} to 0.70(3). A picture of two gluons bound by a short-range attraction, with an unbound 2^{++}, is not killed: the VR 2^{++} lies above the free box level (3.1σ at 2.4, 4.1σ at 2.35). Langfeld and Schafke [13] made the opposite reduction, keeping the vortex part, and found 0^{+} and 2^{+} screening masses within 10% of the full ones; together with ours this suggests that the part left after vortex removal carries a smaller and non-uniform share of the spectrum, a statement limited by our gauge-copy dependence and by the different methods (cooling and screening masses there). The excited 0^{++} levels are upper bounds, so the level structure above the ground state is not settled, and the observation that the VR 2^{++} sits near twice the VR 0^{++} (sec. 4.6) is post hoc.

At $\beta = 2.5$ the vortex-removed mass is also volume independent, and the long-time estimators are consistent with [1, 2] at the largest distances we can resolve ($R \approx 7a$, about 1.3 to $1.5/\sqrt{\sigma}$), but the pre-registered short-time check does not pass, and the VR mass moves at the 2σ level with the gauge copy. The center-projected string tension depends strongly on the gauge copy (0.99 of the full one with a single copy, 0.73 with the best of 16, falling almost monotonically with the functional, as in [14, 15]; at 2.4 it falls from 1.06 to 0.87 with the best of 8); since the VR links are the original links with the center part divided out, the VR ensemble itself depends on the copy, and a VR mass has to be quoted with that systematic. At $\beta = 2.3$ the VR correlator is not a clean exponential at the available times, and at 2.35 the VR effective mass does not plateau either.

What would settle it: VR masses at gauge copies near the global maximum (simulated annealing, many copies) and in Laplacian center gauge, at several couplings and on the same configurations; larger T so that VR correlators can be read beyond $t = 3$; a premise check built on long-time or Creutz-ratio estimators fixed before new data; a third box per coupling; and Polyakov-loop measurements on the VR ensembles.

Data and code

Data tables (CSV) and the simulation and analysis code for all four series accompany this note in the same repository. Run on the original measurement files, the series 1, 2 and 4 analysis scripts reproduce data/series1_summary.csv, data/series2_summary.csv and data/series4_summary.csv exactly. The measurement files (several GB) are not included; the code regenerates statistically equivalent ensembles from the listed seeds.

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